EL-HOUCINE OUSSI

Software Engineer • Machine Learning Engineer

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Moroccan Citizen • M.Sc. in Computer Science at NUIST (Expected June 2027) • Open to Global Relocation & Remote Roles

EDUCATION

Nanjing University of Information Science and Technology (NUIST) — Nanjing, China

Master of Science in Computer Science and Technology — Sept 2024 – June 2027 (Expected)

- Research in solar power forecasting, spatio-temporal deep learning, and atmospheric numerical data modeling.

SUPMTI (Higher School of Management, Telecom & Computer Science) — Oujda, Morocco

State Engineering Degree in Computer Science & Information Systems (5-Year Program) — Sept 2021 – June 2024

- Specialization in Distributed Systems, Software Engineering, Clean Architecture, and Database Systems.

ISTA Lazaret (Specialized Institute of Applied Technology) — Oujda, Morocco

Specialized Technician Diploma in Software Development (TS) — Sept 2019 – June 2021

- Core curriculum in Object-Oriented Programming (OOP), Relational Databases (SQL Server/MySQL), and Web Development.

TECHNICAL SKILLS

Languages:	Python, C# (.NET), Dart, Java, C/C++, TypeScript, JavaScript, SQL, Bash
Frameworks & ML:	ASP.NET Core, Flutter, PyTorch, LightGBM, XGBoost, Scikit-learn, Django, Spring Boot, React
Cloud & Databases:	PostgreSQL, Microsoft SQL Server, MySQL, SQLite, MongoDB, Firebase, Docker, Linux, IIS
Architecture & Tools:	Clean Architecture, RESTful APIs, MVVM (Riverpod), Microservices, Spatio-Temporal Time Series, CI/CD, Git

WORK & RESEARCH EXPERIENCE

NUIST Renewable Energy AI Lab — Nanjing, China

Graduate AI & Data Science Researcher — Sept 2024 – Present

- Architected an end-to-end benchmarking pipeline across 50 commercial Australian solar farms, evaluating persistence baselines, LightGBM, and neural architectures across 15-minute to day-ahead lead times.
- Ingested and aligned multi-terabyte spatio-temporal datasets comprising Himawari-8/9 geostationary satellite channels and ECMWF ERA5 numerical weather fields.
- Authored and released the open-source benchmark dataset and evaluation harness on Zenodo (DOI: 10.5281/zenodo.23590205) with 100% byte-reproducible execution recipes.

Majal Berkane – Smart City — Berkane, Morocco

Backend & Mobile Development Intern — Mar 2024 – July 2024

- Developed a cross-platform mobile application in Flutter using Riverpod for MVVM reactive state management.
- Engineered high-throughput RESTful Web APIs in ASP.NET Core following Clean Architecture and repository patterns.
- Optimized SQL Server query plans and in-memory caching for live IoT sensor feeds, reducing telemetry retrieval latency by 35%.

Direction Provinciale de l'Équipement, du Transport et de la Logistique — Oujda, Morocco

Web Engineering Intern — Mar 2023 – July 2023

- Built an internal administrative platform in Django automating personnel tracking, heavy equipment dispatch, and logistics approval workflows.
- Implemented role-based access control (RBAC) and automated data reporting pipelines using Pandas and NumPy.
- Configured and deployed production application on Windows Server with IIS reverse proxy and automated SQL Server database backups.

SELECTED PROJECTS

PV Forecast Benchmark – Solar Power Forecasting Platform — 2024 – Present

Python, PyTorch, LightGBM, Pandas, Zenodo • Code • DOI: 10.5281/zenodo.23590205

- Built a standardized benchmarking framework assessing forecast models across 50 solar generation sites to resolve accuracy tradeoffs between physical, machine learning, and deep learning models.

Connect Check – Real-Time Interactive Mobile Application — 2024 – 2025

Flutter, Dart, Firebase, Riverpod, SQLite • GitHub

- Engineered a real-time cross-platform mobile application featuring live cloud synchronization with Firebase, dynamic multi-language localization via Riverpod, and offline caching with SQLite.

EcoBeach – Coastal Environmental Monitoring Platform — 2024

ASP.NET Core 8, C#, Clean Architecture, IoT, SQL Server

- Designed a Clean Architecture enterprise backend for coastal water quality and ecological monitoring, integrating IoT sensors, alert thresholds, and analytical reporting.

MystoGuess – Mobile Puzzle Game (Google Play Store) — 2024

Flutter, Dart • Published on Google Play • GitHub

- Published mobile game on Google Play Store featuring dynamic animations, audio engine, and offline state persistence.

LANGUAGES

English: Fluent (Professional working & scientific research) • **French:** Proficient (Professional working proficiency) • **Chinese (Mandarin):** Conversational (HSK 3) • **Arabic:** Native • **Tamazight:** Native